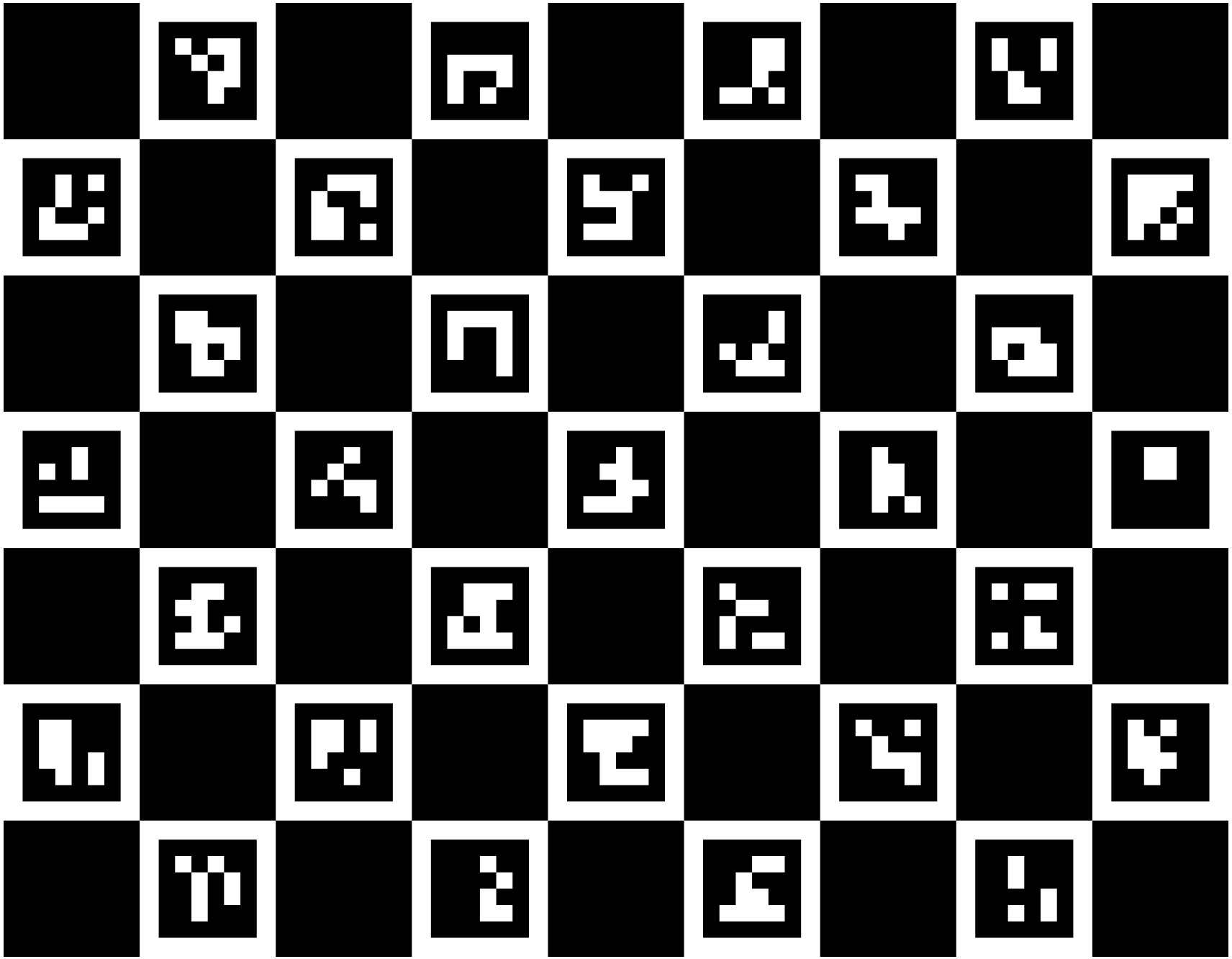
100.00 mm scale bar

Print at 100% (no fit-to-page). Then measure the printed square and record it.



ChArUco 9x7 | nominal square 25.00 mm | marker 18.00 mm | DICT_4X4_50 | inner corners 48

ChArUco capture protocol

1. Print and mount

Print this page at 100% on matte A4 cardstock. Do not use 'fit to page'.
Glue the sheet flat to a rigid board (foam board, MDF or glass). A taped sheet still bows, and warp cannot be corrected later: it shows up as reprojection error that will not come down.
Check the 100 mm bar above: it must measure 100.0 mm +/- 0.3 mm.

2. Measure the print, then record it

Measure three squares horizontally and three vertically with the instrument you will use for the object, and enter the means:

```
python -m calibration.make_board measure --spec calibration/board/board_spec.json \
    --square-x-mm <mean> --square-y-mm <mean> --instrument "<model, 0.01 mm, +/-0.02 mm>"
```

If the two axes differ by more than 0.3% the print is stretched: reprint.

3. Lock the camera (Pixel 8 Pro)

Main (wide) lens only. Disable automatic lens switching, ultra-wide and telephoto.
Zoom fixed at 1.0x. Lock focus at the working distance and never change it again.
Lock ISO and white balance. Disable HDR, night mode and portrait mode.
4:3, maximum JPEG quality, no HEIC. Keep one phone orientation for every photo.

4. Capture 30 or more views

Move the board, not the camera: tilt +/-30 degrees around both axes, rotate it in plane, and place it centre, corners and edges of the frame.
Vary distance so the board fills 40-70% of the frame. Reject blurred frames.
Diffuse daylight only: no direct lamp reflection on the board.
Every view needs at least 12 inner corners, 20+ is better. Never crop the images.

5. Transfer and calibrate

Copy the original JPEGs over USB/MTP (not a messaging app) into calibration/images/.
The command applies EXIF rotation and resizes to the 2048x1536 working frame.
Run: `python -m calibration.calibrate --images calibration/images`
Read calibration/report/calibration_report.md, then look at undistort_check.png.

Measurement record (write it down before calibrating)

Square size, horizontal mean (mm): .....
Square size, vertical mean (mm): .....
Instrument, resolution, accuracy: .....
Date and initials: .....